

June 30, 2026 09:51 AM GMT

Electronic Components | Japan

May HDD/SSD Data: DC Volumes Remain High

Key Takeaways

- DC storage likely to stay segmented between SSDs, HDDs, and tape drives.
- TSR forecasts 2026 DC-use enterprise SSD shipment capacity at 509.33EB, up 94.0% YoY.
- TSR forecast 2026 DC-use NL HDD shipment capacity at 1,841EB, up 30.1% YoY.

No change to view for continued growth in total shipment capacity of both NL HDDs and enterprise SSDs for data center use: We continue to expect sustained growth in total shipment capacity for both NL HDDs and enterprise SSDs, driven by ongoing high production levels and rising capacity per unit. TSR slightly lowered its 2026 HDD shipment capacity forecast to 2,055.5EB (+26.3% YoY) from 2,058.2EB (+26.4%) last month, but still expects continued growth from 1,627.9EB in 2025 (+29.5%). Meanwhile, TSR raised its 2026 total SSD shipment capacity forecast to 759.36EB (+52.5% YoY) from 675.30EB (+35.6%) last month, compared with 498.0EB in 2025 (+38.0%). With higher-capacity products accounting for an increasing share of shipments for both NL HDDs and enterprise SSDs, we expect shipment capacity growth to continue outpacing unit shipment growth.

May DC NL HDD output +14.8% YoY, +4.0% MoM: May NL HDD production was 7.07mn units (6.16mn a year ago, 6.80mn a month ago). NL output declined from 81.45mn units in 2021 (+18.0% YoY) to 71.30mn in 2022 (-12.5%) and 44.51mn in 2023 (-37.6%), before recovering to 63.12mn in 2024 (+41.8%) and 74.77mn in 2025 (+18.5%). TSR forecasts 84.37mn units for 2026 (+12.8% YoY), up from last month's forecast of 83.97mn (+12.3% YoY). It estimates Jan-Mar production of 20.77mn units (+19.9% YoY, +5.0% QoQ) and Apr-Jun production of 20.97mn units (+14.2% YoY, +1.0% QoQ).

May DC-use enterprise SSD shipments (capacity) +138.8% YoY, +9.7% MoM: May enterprise SSD shipment capacity totaled 41.35EB (17.32EB a year ago, 37.70EB a month ago). Full-year shipment capacity was 77.47EB in 2021, 109.83EB (+41.8% YoY) in 2022, 102.66EB (-6.5%) in 2023, 165.95EB (+61.6%) in 2024, and 262.5EB (+58.2% YoY) in 2025. TSR forecasts 2026 enterprise SSD shipment capacity at 509.33EB (+94.0% YoY), up from last month's forecast of 476.1EB (+62.0% YoY).

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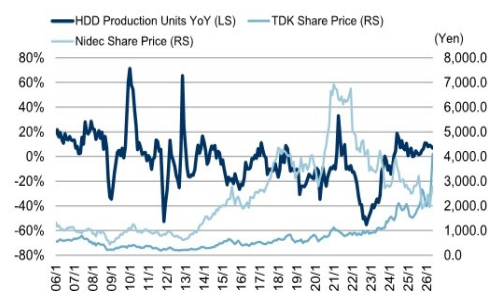
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Industry View

In-Line

Exhibit 1 : HDD Production Volume YoY and Share Prices of TDK & Nidec



Source: TSR, Morgan Stanley Research

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May HDD output +6.9% YoY, +0.7% MoM: May HDD production was 11.26mn units (10.54mn a year ago, 11.18mn a month ago). HDD output declined from 261.70mn units in 2021 (–0.1% YoY) to 168.70mn in 2022 (–35.5%), then to 120.51mn in 2023 (–28.6%), but rose to 124.77mn in 2024 (+3.5%) and to 129.54mn in 2025 (+3.8%). TSR forecasts 134.85mn units for 2026 (+4.1% YoY), slightly below last month's forecast of 135.04mn (+4.2% YoY). It estimates Jan–Mar production at 33.89mn units (+12.9% YoY, +0.1% QoQ) and Apr–Jun production at 33.76mn units (+6.6% YoY, –0.4% QoQ), versus last month's forecast of 33.81mn (+6.7% YoY, –0.2% QoQ). Our forecasts shown in Exhibit 4 onward reflect shipments and therefore differ from TSR's production estimates.

May SSD shipments –0.5% YoY, +1.5% MoM: May SSD shipments were 29.64mn units (29.80mn a year ago, 29.23mn a month ago). SSD shipments declined from 383.02mn units in 2021 (+21.4% YoY) to 348.30mn in 2022 (–9.1%), then to 319.80mn in 2023 (–8.2%), but recovered to 348.20mn in 2024 (+8.0%) and to 375.01mn in 2025 (+7.7%). TSR forecasts shipments of 380.46mn units for 2026 (+1.5% YoY), compared with last month's forecast of 363.90mn (–3.0% YoY). It estimates Jan–Mar shipments at 89.20mn units (+2.6% YoY, –11.4% QoQ) and Apr–Jun shipments at 89.77mn units (flat YoY, +0.6% QoQ), versus last month's forecast of 90.38mn (+0.6% YoY, +1.3% QoQ).

May DC-use enterprise SSD shipments +45.8% YoY, +2.6% MoM: May enterprise SSD shipments were 5.83mn units (4.00mn a year ago, 5.68mn a month ago). Enterprise SSD shipments rose from 36.10mn units in 2021 (+6.8% YoY) to 47.30mn in 2022 (+31.0%), then fell to 40.10mn in 2023 (–15.2%), before recovering to 50.00mn in 2024 (+24.7%) and to 56.51mn in 2025 (+13.1%). TSR forecasts 72.31mn units for 2026 (+28.0% YoY), up from last month's forecast of 71.53mn (+26.6% YoY). It estimates Jan–Mar shipments at 17.20mn units (+41.0% YoY, +2.9% QoQ) and Apr–Jun shipments at 17.66mn units (+41.3% YoY, +2.7% QoQ), compared with last month's forecast of 17.63mn units.

Exhibit 2: HDD Production Volume by Company and Size

(k units)		25/5	25/6	25/7	25/8	25/9	25/10	25/11	25/12	26/1	26/2	26/3	26/4	26/5	YoY	MoM
Seagate	Enterprise	150	150	150	150	150	150	100	160	110	110	160	140	130	-13.3%	-7.1%
	NL	2,500	2,600	2,600	2,650	2,650	2,700	2,800	2,900	2,850	2,870	2,700	2,800	2,850	14.0%	1.8%
	3.5"ATA	1,050	900	1,000	1,000	1,000	1,000	900	1,100	1,000	950	900	950	950	-9.5%	0.0%
	2.5"Mobile	600	700	700	650	650	650	650	650	600	550	600	550	500	-16.7%	-9.1%
	Total	4,300	4,350	4,450	4,450	4,450	4,500	4,450	4,810	4,560	4,480	4,360	4,440	4,430	3.0%	-0.2%
WDC	Enterprise															
	NL	2,900	2,900	2,900	2,950	2,950	2,960	3,160	3,080	3,150	3,200	3,100	3,200	3,350	15.5%	4.7%
	3.5"ATA	1,000	1,100	1,200	1,150	1,150	1,100	1,050	1,100	1,150	1,100	1,100	1,050	1,000	0.0%	-4.8%
	2.5"Mobile	600	650	650	700	650	600	600	600	580	580	650	600	550	-8.3%	-8.3%
	Total	4,500	4,650	4,750	4,800	4,750	4,660	4,810	4,780	4,880	4,880	4,850	4,850	4,900	8.9%	1.0%
Toshiba	Enterprise	140	130	130	110	100	140	160	140	120	90	100	60	70	-50.0%	16.7%
	NL	760	760	720	770	770	920	870	730	880	980	1,040	800	870	14.5%	8.7%
	3.5"ATA	220	300	240	210	350	240	415	330	320	270	500	330	310	40.9%	-6.1%
	2.5"Mobile	620	630	660	660	640	680	660	540	500	480	600	700	680	9.7%	-2.9%
	Total	1,740	1,820	1,750	1,750	1,860	1,980	2,105	1,740	1,820	1,820	2,240	1,890	1,930	10.9%	2.1%
Total	Enterprise	290	280	280	260	250	290	260	300	230	200	260	200	200	-31.0%	0.0%
	NL	6,160	6,260	6,220	6,370	6,370	6,580	6,830	6,710	6,880	7,050	6,840	6,800	7,070	14.8%	4.0%
	(Air)	1,310	1,190	1,170	1,245	1,295	1,270	1,280	1,280	1,220	1,210	1,120	1,070	1,130	-13.7%	5.6%
	(He)	4,850	5,070	5,050	5,125	5,075	5,310	5,550	5,430	5,660	5,840	5,720	5,730	5,940	22.5%	3.7%
	3.5"ATA	2,270	2,300	2,440	2,360	2,500	2,340	2,365	2,530	2,470	2,320	2,500	2,330	2,260	-0.4%	-3.0%
	2.5"Mobile	1,820	1,980	2,010	2,010	1,940	1,930	1,910	1,790	1,680	1,610	1,850	1,850	1,730	-4.9%	-6.5%
	Total	10,540	10,820	10,950	11,000	11,060	11,140	11,365	11,330	11,260	11,180	11,450	11,180	11,260	6.8%	0.7%
YoY % Change	Enterprise	-9.4%	-12.5%	-15.2%	-27.8%	-21.9%	-17.1%	-27.8%	-9.1%	-23.3%	-42.9%	-13.3%	-28.6%	-31.0%		
	NL	17.3%	19.7%	13.7%	11.8%	11.2%	15.4%	20.2%	14.5%	22.0%	20.5%	17.3%	14.3%	14.8%		
	(Air)	-5.8%	-11.2%	-24.5%	-14.1%	-15.6%	-17.8%	-16.1%	-15.0%	-19.2%	-18.2%	-11.1%	-21.3%	-13.7%		
	(He)	25.6%	30.3%	28.8%	20.6%	21.0%	27.8%	33.6%	24.7%	37.0%	33.6%	25.2%	24.8%	22.5%		
	3.5"ATA	-17.5%	-14.2%	-11.9%	-9.9%	-9.1%	-4.9%	-8.0%	15.5%	5.1%	0.9%	1.2%	2.2%	-0.4%		
	2.5"	-15.7%	-6.6%	-7.8%	-6.5%	-4.0%	-8.1%	-8.2%	-2.2%	-10.2%	-13.4%	-2.1%	2.2%	-4.9%		
	Total	0.6%	4.5%	2.8%	1.6%	2.2%	5.0%	6.3%	11.0%	10.8%	7.9%	9.2%	8.3%	6.8%		
MoM % Change	Enterprise	3.6%	-3.4%	0.0%	-7.1%	-3.8%	16.0%	-10.3%	15.4%	-23.3%	-13.0%	30.0%	-23.1%	0.0%		
	NL	3.5%	1.6%	-0.6%	2.4%	0.0%	3.3%	3.8%	-1.8%	2.5%	2.5%	-3.0%	-0.6%	4.0%		
	(Air)	-3.7%	-9.2%	-1.7%	6.4%	4.0%	-1.9%	0.8%	0.0%	-4.7%	-0.8%	-7.4%	-4.5%	5.6%		
	(He)	5.7%	4.5%	-0.4%	1.5%	-1.0%	4.6%	4.5%	-2.2%	4.2%	3.2%	-2.1%	0.2%	3.7%		
	3.5"ATA	-0.4%	1.3%	6.1%	-3.3%	5.9%	-6.4%	1.1%	7.0%	-2.4%	-6.1%	7.8%	-6.8%	-3.0%		
	2.5"	0.6%	8.8%	1.5%	0.0%	-3.5%	-0.5%	-1.0%	-6.3%	-6.1%	-4.2%	14.9%	0.0%	-6.5%		
	Total	2.1%	2.7%	1.2%	0.5%	0.5%	0.7%	2.0%	-0.3%	-0.6%	-0.7%	2.4%	-2.4%	0.7%		

Source: TSR, Morgan Stanley Research

Exhibit 3: SSD Shipments/Capacity by Application

(k units)		25/5	25/6	25/7	25/8	25/9	25/10	25/11	25/12	26/1	26/2	26/3	26/4	26/5	YoY	MoM
SSD shipment	Enterprise	4,000	4,500	4,870	5,010	5,220	5,500	5,610	5,600	5,630	5,670	5,900	5,680	5,830	45.8%	2.6%
	PC	18,500	19,500	19,000	20,000	20,500	21,000	21,500	19,000	18,000	17,000	17,500	17,430	17,500	-5.4%	0.4%
	Add-on	4,100	4,200	4,200	4,300	4,500	4,500	4,600	4,400	4,000	3,700	3,500	3,560	3,630	-11.5%	2.0%
	Game	2,400	2,600	2,000	2,500	3,000	2,500	2,000	2,000	1,800	1,800	1,800	1,600	1,700	-29.2%	6.3%
	Others	800	900	800	800	900	800	800	900	950	980	970	960	980	22.5%	2.1%
	Total	29,800	31,700	30,870	32,610	34,120	34,300	34,510	31,900	30,380	29,150	29,670	29,230	29,640	-0.5%	1.4%
YoY % Change	Enterprise	-4.8%	-6.3%	13.3%	11.3%	8.7%	10.0%	19.4%	24.4%	34.0%	35.0%	55.3%	42.0%	45.8%		
	PC	1.1%	5.4%	4.4%	8.1%	9.6%	10.5%	16.2%	9.2%	5.9%	-5.6%	-7.9%	-0.4%	-5.4%		
	Add-on	10.8%	5.0%	13.5%	13.2%	15.4%	12.5%	12.2%	2.3%	2.6%	-7.5%	-14.6%	-11.0%	-11.5%		
	Game	20.0%	30.0%	0.0%	25.0%	50.0%	25.0%	0.0%	66.7%	-5.3%	-10.0%	-25.0%	-20.0%	-29.2%		
	Others	14.3%	12.5%	14.3%	0.0%	12.5%	0.0%	0.0%	12.5%	18.8%	22.5%	21.3%	20.0%	22.5%		
	Total	3.1%	5.3%	6.8%	10.2%	13.0%	11.4%	14.7%	13.1%	9.3%	0.5%	-1.4%	3.3%	-0.5%		
MoM % Change	Enterprise	0.0%	12.5%	8.2%	2.9%	4.2%	5.4%	2.0%	-0.2%	0.5%	0.7%	4.1%	-3.7%	2.6%		
	PC	5.7%	5.4%	-2.6%	5.3%	2.5%	2.4%	2.4%	-11.6%	-5.3%	-5.6%	2.9%	-0.4%	0.4%		
	Add-on	2.5%	2.4%	0.0%	2.4%	4.7%	0.0%	2.2%	-4.3%	-9.1%	-7.5%	-5.4%	1.7%	2.0%		
	Game	20.0%	8.3%	-23.1%	25.0%	20.0%	-16.7%	-20.0%	0.0%	-10.0%	0.0%	0.0%	-11.1%	6.3%		
	Others	0.0%	12.5%	-11.1%	0.0%	12.5%	-11.1%	0.0%	12.5%	5.6%	3.2%	-1.0%	-1.0%	2.1%		
	Total	5.3%	6.4%	-2.6%	5.6%	4.6%	0.5%	0.6%	-7.6%	-4.8%	-4.0%	1.8%	-1.5%	1.4%		
SSD capacity (EB)	Enterprise	17.32	20.64	22.40	23.51	24.60	28.02	28.88	30.63	32.91	35.03	36.80	37.70	41.35	138.8%	9.7%
	PC	11.71	12.85	12.85	13.62	14.06	14.55	14.94	13.44	12.93	12.22	12.61	12.60	12.67	8.2%	0.6%
	Add-on	3.55	3.93	3.86	4.16	4.41	4.55	4.84	4.54	4.37	3.96	3.52	3.73	3.83	8.0%	2.8%
	Game	2.80	3.20	2.50	3.00	3.50	3.00	2.40	2.30	2.30	2.20	2.20	1.90	2.00	-28.6%	5.3%
	Others	0.11	0.14	0.13	0.13	0.16	0.14	0.15	0.17	0.19	0.20	0.20	0.21	0.22	93.5%	2.1%
	Total	35.49	40.76	41.74	44.41	46.72	50.27	51.20	51.07	52.69	53.60	55.33	56.15	60.08	69.3%	7.0%

Source: TSR, Morgan Stanley Research

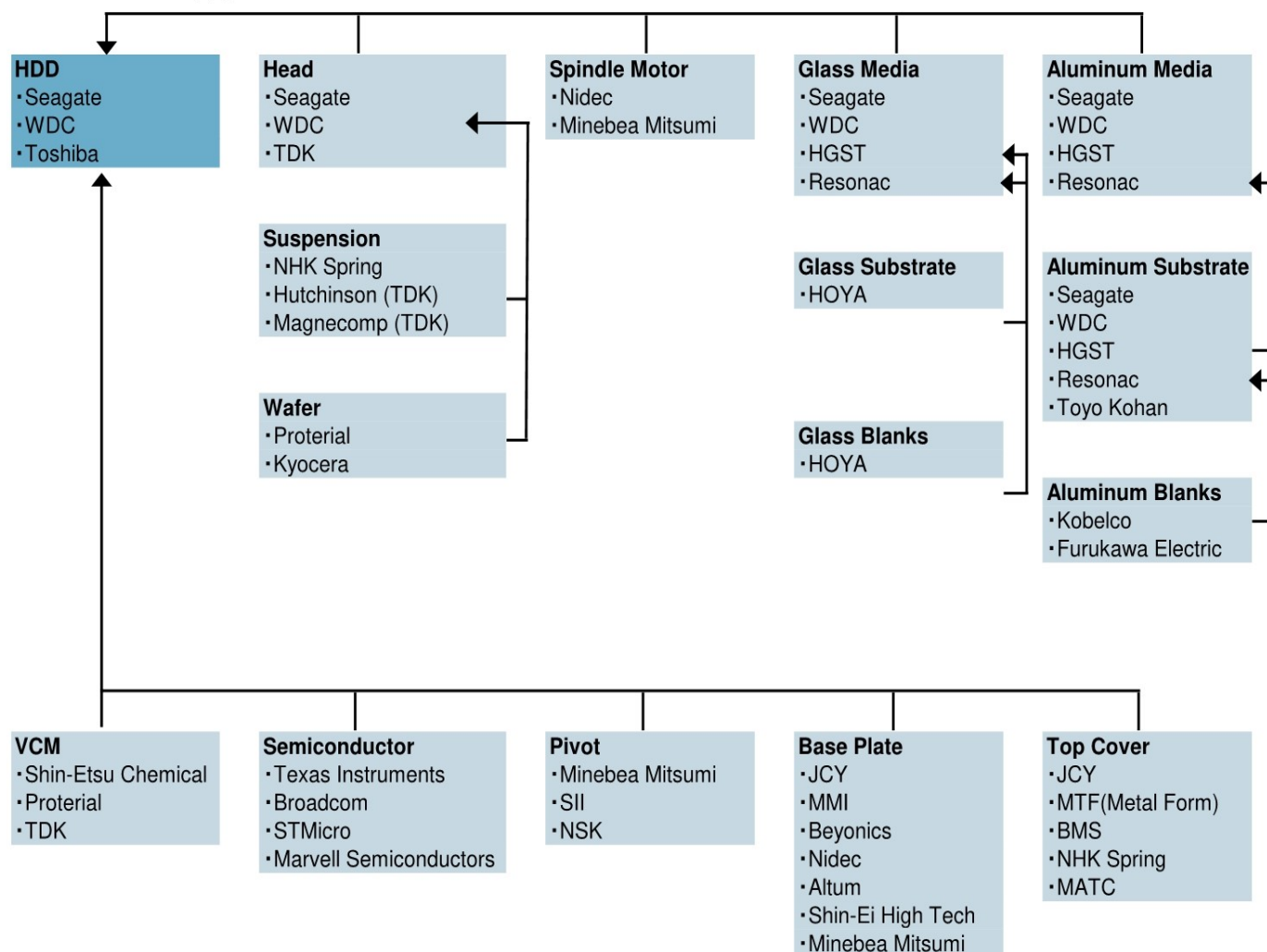
Exhibit 4: HDD Shipment Volume / Shipment Capacity

	CY23				CY24				CY25				CY26					CY14	CY15	CY16	CY17	CY18	CY19	CY20	CY21	CY22	CY23	CY24	CY25	CY26e	CY27e	CY28e
	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q																
HDD Shipment (by Size)																																
Total	33.5	31.3	28.6	28.9	29.7	30.3	32.1	31.7	28.7	30.4	31.6	33.0	33.0		564.1	468.8	423.9	403.1	375.4	317.7	260.0	259.3	172.2	122.3	123.9	123.8	126.8	125.7	125.3			
Enterprise	1.4	1.6	1.4	1.4	1.5	1.0	1.0	0.9	0.8	0.8	0.7	0.8	0.8		32.4	32.1	24.8	22.6	21.5	17.4	12.9	13.4	8.4	5.8	4.4	3.2	2.3	1.5	0.8			
NL	11.4	9.2	9.0	9.9	12.3	13.8	15.9	16.5	14.6	16.5	17.3	19.2	18.7		36.9	37.2	41.0	42.4	51.3	52.7	59.1	73.5	61.5	39.4	58.6	67.6	75.5	79.6	81.3			
3.5"ATA	11.2	12.1	11.2	10.7	9.3	9.4	9.0	7.9	7.6	7.9	8.0	7.6	8.3		202.6	161.0	142.9	135.4	120.5	103.9	87.0	89.9	55.0	45.2	35.6	30.9	28.8	27.6	26.8			
2.5"	9.5	8.4	7.1	6.9	6.6	6.1	6.2	6.4	5.7	5.3	5.6	5.5	5.3		291.7	238.6	215.1	202.7	182.0	143.6	101.0	82.5	46.3	32.0	25.3	22.0	20.3	17.1	16.4			
1.8"	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
YoY % Change	-37%	-30%	-26%	-20%	-11%	-3%	12%	10%	-3%	0%	-2%	4%	15%		2%	-17%	-10%	-5%	-7%	-15%	-18%	0%	-33.6%	-29%	1%	0%	2%	-1%	0%			
Enterprise	-48%	-34%	-15%	-16%	6%	-36%	-30%	-33%	-45%	-21%	-23%	-16%	-7%		2%	-1%	-23%	-9%	-5%	-19%	-26%	3%	-37%	-32%	-23%	-29%	-28%	-36%	-47%			
NL	-39%	-49%	-37%	-7%	8%	51%	78%	67%	19%	19%	9%	16%	28%		4%	1%	10%	3%	21%	3%	12%	24%	-16.3%	-36%	49%	15.5%	11.6%	5.4%	2.2%			
3.5"ATA	-33%	-4%	-14%	-16%	-16%	-22%	-19%	-26%	-19%	-17%	-12%	-4%	9%		0%	-21%	-11%	-5%	-11%	-14%	-16%	3%	-39%	-18%	-21%	-13%	-7%	-4%	-3%			
2.5"	-36%	-20%	-26%	-39%	-30%	-28%	-13%	-7%	-14%	-13%	-10%	-14%	-7%		4%	-18%	-10%	-6%	-10%	-21%	-30%	-18%	-44%	-31%	-21%	-13%	-8%	-16%	-4%			
1.8"	-	-	-	-	-	-	-	-	-	-	-	-	-		-6%	-98%	-	-	-	-	-	-	-	-	-	-	-	-	-			
Sequential % Change	-8%	-7%	-9%	1%	3%	2%	6%	-1%	-10%	6%	4%	5%	0%		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
Enterprise	-16%	13%	-16%	4%	7%	-32%	-7%	-1%	-13%	-1%	-10%	8%	-4%		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
NL	8%	-20%	-2%	11%	24%	12%	15%	4%	-12%	13%	5%	11%	-3%		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
3.5"ATA	-13%	9%	-7%	-4%	-13%	1%	-4%	-13%	-4%	2%	-5%	10%	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
2.5"	-16%	-11%	-16%	-3%	-4%	-8%	2%	3%	-11%	-8%	6%	-1%	-3%		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
1.8"	-	-	-	-	-	-	-	-	-	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
HDD Shipment (by Company)																																
Total	33.5	31.3	28.6	28.9	29.7	30.3	32.2	31.7	28.7	30.4	31.6	33.0	33.0		564.1	468.8	423.9	403.1	375.4	317.7	260.0	259.3	172.2	122.3	123.9	123.8	126.8	125.7	125.3			
Seagate	14.5	14.3	12.0	11.6	11.4	12.3	13.0	13.0	11.4	12.5	12.5	13.1	13.1		224.1	188.6	155.2	148.2	150.0	129.1	126.7	110.3	74.1	52.4	49.7	49.4	-	-	-	-		
Western Digital	12.6	11.8	10.4	10.8	11.7	12.1	13.2	13.5	12.1	12.9	13.9	14.3	14.4		249.3	215.6	191.4	161.8	139.7	114.2	110.8	94.3	63.9	45.6	50.5	53.2	-	-	-	-		
Toshiba	6.4	5.2	6.2	6.5	6.7	5.9	6.0	5.2	5.2	5.0	5.2	5.7	5.6		90.7	75.8	93.5	93.0	85.7	74.4	70.0	54.7	34.3	24.3	23.8	21.1	-	-	-	-		
Market Share																																
Seagate	43%	46%	42%	40%	39%	41%	40%	41%	40%	41%	39%	39%	40%		40%	40%	37%	37%	40%	41%	49%	43%	43%	43%	40%	40%	-	-	-	-		
Western Digital	38%	38%	36%	37%	39%	40%	41%	43%	42%	42%	44%	43%	44%		44%	46%	45%	40%	37%	36%	43%	36%	37%	37%	41%	43%	-	-	-	-		
Toshiba	19%	17%	22%	22%	23%	20%	19%	17%	18%	16%	17%	17%	17%		16%	16%	22%	23%	23%	23%	27%	21%	20%	20%	19%	17%	-	-	-	-		
YoY % Change	-37%	-30%	-26%	-20%	-11%	-3%	12%	10%	-3%	0%	-2%	4%	15%		2%	-17%	-10%	-5%	-7%	-15%	-18%	0%	-34%	-29%	1%	0%	2%	-1%	0%			
Seagate	-37%	-28%	-24%	-25%	-21%	-14%	8%	12%	0%	2%	-4%	0%	15%		1%	-16%	-18%	-4%	1%	-14%	-2%	-13%	-33%	-29%	-5%	0%	-	-	-	-		
Western Digital	-36%	-28%	-29%	-16%	-7%	3%	27%	25%	3%	7%	5%	6%	19%		1%	-14%	-11%	-15%	-14%	-18%	-3%	-15%	-32%	-29%	11%	5%	-	-	-	-		
Toshiba	-37%	-37%	-22%	-19%	6%	14%	-4%	-19%	-22%	-16%	-12%	8%	7%		8%	-16%	23%	-1%	-8%	-13%	-6%	-22%	-37%	-29%	-2%	-11%	-	-	-	-		
Sequential % Change	-8%	-7%	-9%	1%	3%	2%	6%	-1%	-10%	6%	4%	5%	0%		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
Seagate	-6%	-1%	-16%	-3%	-2%	8%	5%	0%	-12%	10%	0%	5%	0%		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
Western Digital	-2%	-6%	-12%	4%	8%	3%	9%	2%	-10%	7%	8%	3%	0%		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
Toshiba	-21%	-19%	20%	5%	3%	-11%	0%	-12%	-1%	-4%	5%	8%	-2%		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
HDD Shipment Capacity (EB, by Company)																																
Total	236.0	189.8	196.1	214.3	264.6	289.4	340.3	362.3	346.6	391.8	432.5	457.0	477.2		534.9	535.8	622.6	720.7	855.3	902.5	1,019.1	1,344.8	1,163.5	836.2	1,256.6	1,627.9	2,004.5	2,463.5	3,027.8			
Seagate	118.7	91.2	89.6	95.1	99.1	114.2	137.5	150.8	143.6	162.5	181.7	190.1	199.0		221.6	222.7	252.2	287.2	366.5	367.0	480.6	614.2	539.5	394.6	501.6	677.9	-	-	-	-		
Western Digital	96.5	78.8	79.1	90.3	127.3	142.6	163.8	172.5	162.0	187.0	204.1	215.0	222.0		240.1	250.1	286.1	342.7	390.0	422.7	422.3	544.0	476.3	344.7	606.2	768.1	-	-	-	-		
Toshiba	20.8	19.8	27.4	28.9	38.2	32.6	39.2	39.0	41.0	42.3	46.7	51.9	56.1		73.2	63.0	84.3	90.8	98.8	112.9	116.2	187.5	147.7	96.9	149.0	181.9	-	-	-	-		
YoY % Change	-28%	-41%	-31%	-6%	12%	52%	74%	69%	31%	35%	27%	26%	38%		16%	0%	16%	16%	19%	6%	13%	32%	-13%	-28%	50%	30%	23%	23%	23%			
Seagate	-23%	-41%	-24%	-15%	-17%	25%	53%	59%	45%	42%	32%	26%	39%		14%	0%	13%	14%	28%	0%	31%	28%	-12%	-27%	27%	35%	-	-	-	-		
Western Digital	-27%	-39%	-39%	8%	32%	81%	107%	91%	27%	31%	25%	25%	37%		19%	4%	14%	20%	14%	8%	0%	29%	-12%	-28%	76%	27%	-	-	-	-		
Toshiba	-51%	-45%	-26%	-10%	64%	65%	43%	35%	7%	30%	19%	33%	37%		14%	-14%	34%	8%	9%	14%	3%	61%	-21%	-34%	54%	22%	-	-	-	-		
Sequential % Change	3%	-20%	3%	9%	23%	9%	18%	6%	-4%	13%	10%	6%	4%		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
Seagate	6%	-23%	-2%	6%	4%	15%	20%	10%	-5%	13%	12%	5%	5%		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
Western Digital	15%	-18%	0%	14%	41%	12%	15%	5%	-6%	15%	9%	5%	3%		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
Toshiba	-35%	-5%	38%	5%	32%	-15%	20%	-1%	5%	3%	10%	11%	8%		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		

Note: e = Morgan Stanley Research estimates

Source: TSR, Morgan Stanley Research

Exhibit 5: HDD Supply Chain



Source: TSR, Morgan Stanley Research

Exhibit 6: Market Shares of HDD Parts Suppliers (as of Jan-Mar 2026)

HDD Head		Suppliers			HDD Motor		Suppliers		HD Media		Suppliers			
		Seagate	WDC	TDK			Nidec	Minebea			Seagate	WDC	HGST	Resonac
Customers	Seagate	99.9%		0.1%	Customers	Seagate	100%		Customers	Seagate	100%			
	Enterprise	100%				Enterprise	100%			Enterprise	100%			
	NL	99.9%		0.1%		NL	100%			NL	100%			
	3.5" ATA	100%				3.5" ATA	100%			3.5" ATA	100%			
	2.5"	100%				2.5"	100%			2.5"	100%			
	Samsung					Samsung				Samsung				
	3.5" ATA					3.5" ATA				3.5" ATA				
	2.5"					2.5"				2.5"				
	WDC		100%			WDC	100%			WDC		100%		
	Enterprise					Enterprise				Enterprise				
	NL					NL				NL				
	3.5" ATA		100%			3.5" ATA	100%			3.5" ATA		100%		
	2.5"		100%			2.5"	100%			2.5"		100%		
	HGST		99.6%	0.4%		HGST	100%			HGST			93%	7%
	Enterprise					Enterprise				Enterprise				
	NL		99.6%	0.4%		NL	100%			NL			93%	7%
	3.5" ATA					3.5" ATA				3.5" ATA				
	2.5"					2.5"				2.5"				
	Toshiba			100%		Toshiba	100%			Toshiba				100%
	Enterprise			100%		Enterprise	100%			Enterprise				100%
	NL			100%		NL	100%			NL				100%
	3.5" ATA			100%		3.5" ATA	100%			3.5" ATA				100%
	2.5"			100%		2.5"	100%			2.5"				100%
	1.8"					1.8"				1.8"				
	Total	41%	46%	13%		Total	60%	40%		Total	42%	4%	40%	15%
	Enterprise	65%		35%		Enterprise	49%	51%		Enterprise	63%			37%
	NL	42%	47%	12%		NL	58%	42%		NL	42%		44%	14%
	3.5" ATA	37%	47%	16%		3.5" ATA	64%	36%		3.5" ATA	38%	50%		12%
	2.5"	33%	38%	29%		2.5"	66%	34%		2.5"	35%	40%		25%

Source: TSR, Morgan Stanley Research

Exhibit 7: HDD Head Shipment Volume

(mn units)	CY23				CY24				CY25				CY26																					
	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	CY14	CY15	CY16	CY17	CY18	CY19	CY20	CY21	CY22	CY23	CY24	CY25						
HDD Head Shipment																																		
Total	262.2	202.9	217.1	237.7	262.5	297.1	327.3	332.4	330.0	348.7	377.6	397.8	401.9				1,931.6	1,756.1	1,702.8	1,714.4	1,715.4	1,496.2	1,517.9	1,747.9	1,349.0	919.9	1,219.3	1,454.1						
TDK	39.6	26.4	35.0	40.3	43.3	47.4	49.3	43.4	47.9	45.2	49.7	51.0	52.8				471.6	376.0	375.0	361.9	330.2	299.6	254.4	338.2	230.8	141.3	183.4	193.8						
Seagate	118.8	92.1	86.9	101.7	103.3	116.1	132.8	140.7	128.3	140.8	155.1	165.2	165.4				634.0	616.2	617.1	636.6	683.5	594.1	652.2	755.2	579.1	399.5	492.9	589.4						
Western Digital	103.8	84.4	95.2	95.7	115.9	133.6	145.2	148.3	153.8	162.7	172.8	181.6	183.7				826.0	763.9	710.7	715.9	701.7	602.5	611.3	654.5	539.1	379.1	543.0	670.9						
Market Share																																		
TDK	15%	13%	16%	17%	16%	16%	15%	13%	15%	13%	13%	13%	13%				24%	21%	22%	21%	19%	20%	17%	19%	17%	15%	15%	13%						
Seagate	45%	45%	40%	43%	39%	39%	41%	42%	39%	40%	41%	42%	41%				33%	35%	36%	37%	40%	40%	43%	43%	43%	43%	40%	41%						
Western Digital	40%	42%	44%	40%	44%	45%	44%	45%	47%	47%	46%	46%	46%				43%	43%	42%	42%	41%	40%	40%	37%	40%	41%	45%	46%						
YoY % Change																																		
TDK	-38%	-47%	-33%	9%	0%	46%	51%	40%	26%	17%	15%	20%	22%				0%	-9%	-3%	1%	0%	-13%	1%	15%	-23%	-32%	33%	19%						
Seagate	-45%	-53%	-41%	-5%	9%	80%	41%	8%	11%	-5%	1%	18%	10%				-3%	-20%	0%	-3%	-9%	-9%	-15%	33%	-32%	-39%	30%	6%						
Western Digital	-40%	-47%	-32%	26%	-13%	26%	53%	38%	24%	21%	17%	17%	29%				1%	-3%	0%	3%	7%	-13%	10%	16%	-23%	-31%	23%	20%						
Western Digital	-33%	-44%	-31%	2%	12%	58%	53%	55%	33%	22%	19%	22%	19%				2%	-8%	-7%	1%	-2%	-14%	1%	7%	-18%	-30%	43%	24%						
HDD Head/HDD	7.84	6.49	7.59	8.22	8.83	9.80	10.20	10.47	11.50	11.47	11.95	12.04	12.18				3.42	3.75	4.02	4.25	4.57	4.71	5.84	6.74	7.83	7.52	9.84	11.75						

Source: TSR, Morgan Stanley Research

Exhibit 8: Trends in In-house Production of HDD Heads/Media

(mn units)	CY23			CY24			CY25			CY26			CY14	CY15	CY16	CY17	CY18	CY19	CY20	CY21	CY22	CY23	CY24	CY25	
	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q												
HDD Head																									
Seagate																									
Total demand	123.1	95.8	87.9	102.0	103.4	116.2	132.9	140.7	128.3	140.8	155.1	165.2	165.6	781.6	717.2	664.2	677.2	693.7	650.6	695.3	800.7	607.2	408.8	493.2	589.4
Volume of in-house production	118.8	92.1	86.9	101.7	103.3	116.1	132.8	140.7	128.3	140.8	155.1	165.2	165.4	634.0	616.2	617.1	653.7	646.2	600.3	652.2	755.2	579.1	399.5	492.9	589.4
Shipment volume from TDK	4.3	3.7	1.0	0.3	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.2	147.6	101.0	47.1	23.5	47.5	50.3	43.1	45.5	28.1	9.3	0.3	0.0
Ratio of in-house production	97%	96%	99%	100%	100%	100%	100%	100%	100%	100%	100%	100%	99.9%	81%	86%	93%	97%	93%	92%	94%	94%	95%	98%	100%	100%
WD																									
Total demand	106.7	86.4	95.4	95.7	115.9	133.6	145.2	148.3	153.8	162.7	172.8	181.6	184.4	882.8	797.4	738.8	749.2	676.2	612.1	611.3	659.3	546.7	384.2	543.0	670.9
ex-WD	20.7	18.8	17.2	16.5	15.0	15.0	15.2	14.5	13.8	14.2	15.8	15.6	15.2	526.2	460.8	386.4	395.0	314.4	228.2	216.5	195.5	95.1	73.2	59.7	59.4
ex-HGST	86.0	67.6	78.2	79.2	100.9	118.6	130.0	133.8	140.0	148.5	157.0	166.0	169.2	356.6	336.6	352.4	354.2	361.8	383.9	394.8	463.8	451.6	311.0	483.3	611.5
Volume of in-house production	103.8	84.4	95.2	95.7	115.9	133.6	145.2	148.3	153.8	162.7	172.8	181.6	183.7	826.0	763.9	710.7	717.0	654.2	611.2	611.3	654.5	539.1	379.1	543.0	670.9
ex-WD	20.7	18.8	17.2	16.5	15.0	15.0	15.2	14.5	13.8	14.2	15.8	15.6	15.2	482.5	443.3	370.2	385.3	309.4	227.3	216.5	195.5	95.1	73.2	59.7	59.4
ex-HGST	83.1	65.6	78.0	79.2	100.9	118.6	130.0	133.8	140.0	148.5	157.0	166.0	168.5	343.5	320.6	340.5	331.7	344.8	383.9	394.8	459.0	444.0	305.9	483.3	611.5
Shipment volume from TDK	2.9	2.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.7	56.8	33.5	28.1	32.2	22.0	0.9	0.0	4.8	7.6	5.1	0.0	0.0
Ratio of in-house production	97%	98%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	99.6%	94%	96%	96%	96%	97%	100%	100%	99%	99%	99%	100%	100%
Toshiba																									
Total demand	32.4	20.7	33.8	40.0	43.2	47.3	49.2	43.4	47.9	45.2	49.7	51.0	51.9	267.2	242.5	299.8	294.2	252.2	254.1	211.3	287.9	195.1	126.9	183.1	193.8
Volume of in-house production	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shipment volume from TDK	32.4	20.7	33.8	40.0	43.2	47.3	49.2	43.4	47.9	45.2	49.7	51.0	51.9	267.2	242.5	299.8	294.2	252.2	254.1	211.3	287.9	195.1	126.9	183.1	193.8
Ratio of in-house production	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Industry Total																									
Total demand	262.2	202.9	217.1	237.7	262.5	297.1	327.3	332.4	330.0	348.7	377.6	397.8	401.9	1,931.6	1,757.1	1,702.8	1,720.6	1,622.1	1,516.8	1,517.9	1,747.9	1,349.0	1,185.1	1,219.3	1,454.1
Shipment volume from TDK	39.6	26.4	35.0	40.3	43.3	47.4	49.3	43.4	47.9	45.2	49.7	51.0	52.8	471.6	377.0	375.0	349.9	321.7	305.3	254.4	338.2	230.8	198.1	183.4	193.8
Market share of TDK	15%	13%	16%	17%	16%	16%	15%	13%	15%	13%	13%	13%	13%	24%	21%	22%	20%	20%	20%	17%	19%	17%	17%	15%	13%
HDD Media																									
Seagate																									
Total demand	64.1	48.7	44.8	51.8	53.2	59.8	67.0	71.0	67.7	73.9	77.6	82.6	84.0	395.5	359.0	330.6	330.3	363.6	327.1	351.8	404.7	308.7	209.4	251.0	301.8
Volume of in-house production	59.6	45.7	42.8	51.8	53.2	59.8	67.0	71.0	67.7	73.9	77.6	82.6	84.0	283.1	258.7	280.6	301.6	325.1	281.6	306.1	363.8	279.5	199.9	251.0	301.8
Shipment volume from Resonac	4.5	3.0	2.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	61.4	58.0	16.9	0.0	0.0	0.0	9.5	29.3	29.2	9.5	0.0	0.0
Ratio of in-house production	93%	94%	96%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	72%	72%	85%	91%	89%	86%	87%	90%	91%	95%	100%	100%
WD																									
Total demand	55.2	44.0	48.5	48.8	60.1	69.0	73.2	74.8	80.0	84.8	87.4	90.8	93.6	442.9	398.7	369.9	374.8	365.8	305.8	309.4	333.9	276.3	196.5	277.1	343.0
ex-WD	11.1	10.2	9.4	9.2	8.4	8.2	8.2	7.9	7.5	7.9	7.9	7.8	7.8	263.1	230.4	193.2	196.2	171.4	125.0	111.9	102.0	50.5	39.9	32.7	31.1
ex-HGST	44.1	33.8	39.1	39.6	51.7	60.8	65.0	66.9	72.5	76.9	79.5	83.0	85.8	179.8	168.3	176.7	178.6	194.4	180.8	197.9	231.9	225.8	156.6	244.4	311.9
Volume of in-house production	49.7	40.0	45.5	45.8	57.1	64.2	68.2	69.8	75.0	79.8	82.4	85.8	87.4	399.9	368.0	343.9	330.7	339.7	300.9	309.4	332.3	259.2	181.0	259.3	323.0
ex-WD	11.1	10.2	9.4	9.2	8.4	8.2	8.2	7.9	7.5	7.9	7.9	7.8	7.8	233.1	211.9	177.2	182.2	163.4	123.0	111.9	102.0	50.5	39.9	32.7	31.1
ex-HGST	38.6	29.8	36.1	36.6	48.7	56.0	60.0	61.9	67.5	71.9	74.5	78.0	79.6	166.8	156.1	166.7	148.5	176.3	177.9	197.5	230.3	208.7	141.1	226.6	291.9
Shipment volume from Resonac	5.5	4.0	3.0	3.0	3.0	4.8	5.0	5.0	5.0	5.0	5.0	5.0	6.2	43.0	30.7	26.0	44.1	26.1	3.2	0.0	1.6	17.1	15.5	17.8	20.0
Ratio of in-house production	90%	91%	94%	94%	95%	93%	93%	93%	94%	94%	94%	94%	93%	90%	92%	93%	88%	93%	98%	100%	100%	94%	92%	94%	94%
Toshiba																									
Total demand	16.5	11.2	19.0	20.0	23.0	22.4	26.0	25.0	29.2	24.0	26.0	27.4	23.8	135.6	122.0	149.9	152.1	133.0	124.3	107.1	145.8	105.3	66.7	96.4	106.6
Volume of in-house production	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shipment volume from Resonac	16.5	11.2	19.0	20.0	23.0	22.4	26.0	25.0	29.2	24.0	26.0	27.4	23.8	135.6	122.0	149.9	152.1	133.0	124.3	107.1	146.1	105.3	66.7	96.4	106.6
Ratio of in-house production	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Industry Total																									
Total demand	135.8	103.9	112.3	120.6	136.3	151.2	166.2	170.8	176.9	182.7	191.0	200.8	201.4	974.0	879.7	850.4	857.2	862.4	757.2	768.3	884.4	690.3	472.6	624.5	751.4
Shipment volume from Resonac	26.5	18.2	24.0	23.0	26.0	27.2	31.0	30.0	34.2	29.0	31.0	32.4	30.0	240.0	210.7	192.8	196.2	159.1	127.5	116.6	177.0	151.6	91.7	114.2	126.6
Market share of Resonac	20%	18%	21%	19%	19%	18%	19%	18%	19%	16%	16%	16%	15%	25%	24%	23%	23%	18%	17%	15%	20%	22%	19%	18%	17%
Shipment volume from Fuji Elec	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	51.0	42.3	33.1	28.7	38.5	45.5	36.2	11.6	0.0	0.0	0.0	0.0
Market share of Fuji Elec	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	5%	5%	4%	3%	4%	6%	5%	1%	0%	0%	0%	0%
HDD Media/HDD																									
	4.06	3.32	3.92	4.17	4.58	4.99	5.18	5.38	6.16	6.01	6.04	6.08	6.10	1.73	1.88	2.01	2.13	2.30	2.38	2.96	3.41	4.01	3.87	5.04	6.07

Exhibit 9: HDD Motor Shipment Volume

(mn units)	CY23				CY24				CY25				CY26		CY14	CY15	CY16	CY17	CY18	CY19	CY20	CY21	CY22	CY23	CY24	CY25
	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q													
HDD Motor Shipment																										
Total	34.2	27.2	31.6	29.5	30.5	32.7	33.2	32.6	30.8	31.7	32.5	34.3	34.0		575.2	476.5	431.7	407.3	381.6	307.7	271.8	248.8	146.1	122.5	129.0	129.3
Nidec	19.2	13.2	19.1	17.4	18.6	18.9	19.3	19.2	18.8	18.8	20.0	20.2	20.4		408.6	380.3	362.6	341.9	317.6	250.1	210.9	133.7	82.1	68.9	76.0	77.8
Minebea Mitsumi	15.0	14.0	12.5	12.1	11.9	13.8	13.9	13.4	12.0	12.9	12.5	14.1	13.6		70.9	67.7	69.1	65.4	63.8	57.5	60.9	115.1	64.0	53.6	53.0	51.5
Market Share																										
Nidec	56%	49%	60%	59%	61%	58%	58%	59%	61%	59%	62%	59%	60%		71%	80%	84%	84%	83%	81%	78%	54%	56%	56%	59%	60%
Minebea Mitsumi	44%	51%	40%	41%	39%	42%	42%	41%	39%	41%	38%	41%	40%		12%	14%	16%	16%	17%	19%	22%	46%	44%	44%	41%	40%
YoY % Change																										
Nidec	-32%	-27%	2%	8%	-11%	20%	5%	11%	1%	-3%	-2%	5%	10%		3%	-17%	-9%	-6%	-6%	-19%	-12%	-8%	-41%	-16%	5%	0%
Minebea Mitsumi	-25%	-30%	3%	-9%	-3%	43%	1%	10%	1%	-1%	4%	5%	9%		4%	-7%	-5%	-6%	-7%	-21%	-16%	-37%	-39%	-16%	10%	2%
Sequential % Change																										
Nidec	25%	-20%	16%	-7%	3%	7%	2%	-2%	-6%	3%	3%	6%	-1%		-	-	-	-	-	-	-	-	-	-	-	-
Minebea Mitsumi	0%	-31%	45%	-9%	7%	2%	2%	-1%	-2%	0%	6%	1%	1%		-	-	-	-	-	-	-	-	-	-	-	-
HDD Motor/HDD	1.02	0.87	1.10	1.02	1.03	1.08	1.03	1.03	1.07	1.04	1.03	1.04	1.03		1.02	1.02	1.02	1.01	1.02	0.97	1.05	0.96	0.85	1.00	1.04	1.04

Source: TSR, Morgan Stanley Research

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	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
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Equal-weight/Hold	1571	43%	369	40%	23%	723	44%
Not-Rated/Hold	3	0%	0	0%	0%	1	0%
Underweight/Sell	551	15%	86	9%	16%	201	12%
Total	3,667		920			1632	

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INDUSTRY COVERAGE: Electronic Components

COMPANY (TICKER)	RATING (AS OF)	PRICE* (06/30/2026)
Shoji Sato		
ALPS ALPINE (6770.T)	O (03/17/2026)	¥2,045
Hamamatsu Photonics (6965.T)	E (06/17/2026)	¥2,671
Ibiden (4062.T)	U (02/04/2026)	¥23,815
Kyocera (6971.T)	E (06/25/2020)	¥3,561
Mabuchi Motor (6592.T)	E (11/03/2022)	¥1,570
Minebea Mitsumi (6479.T)	E (10/23/2025)	¥4,756
Murata Manufacturing (6981.T)	O (11/26/2025)	¥11,395
Nidec (6594.T)	NR (09/05/2025)	¥2,645
Niterra (5334.T)	O (01/17/2024)	¥10,705
Taiyo Yuden (6976.T)	U (06/17/2026)	¥20,150
TDK (6762.T)	O (08/02/2022)	¥3,570
Sota Harashima		
CMK (6958.T)	E (02/28/2025)	¥653
Daishinku (6962.T)	U (06/17/2026)	¥880
Hirose Electric (6806.T)	O (07/10/2024)	¥28,835
IRISO Electronics (6908.T)	E (08/02/2022)	¥2,914
Japan Aviation Electronics Industry (6807.T)	E (01/17/2024)	¥2,311
KOA (6999.T)	E (06/17/2026)	¥2,898
Meiko Electronics (6787.T)	E (04/03/2026)	¥31,000
Nichicon (6996.T)	E (11/10/2021)	¥4,150
Nihon Dempa Kogyo (6779.T)	E (03/07/2024)	¥3,425
Nippon Chemi-Con (6997.T)	U (09/20/2024)	¥5,490

Stock Ratings are subject to change. Please see latest research for each company.

* Historical prices are not split adjusted.